

What Is Bayesian Classification?

Naive Bayes

- ❑ A statistical classifier
 - ❑ Perform *probabilistic prediction* (i.e., predict class membership probabilities)
- ❑ Foundation—Based on Bayes' Theorem
- ❑ Performance
 - ❑ A simple Bayesian classifier, *naïve Bayesian classifier*, has comparable performance with decision tree and selected neural network classifiers
- ❑ Incremental
 - ❑ Each training example can incrementally increase/decrease the probability that a hypothesis is correct—prior knowledge can be combined with observed data
- ❑ Theoretical Standard
 - ❑ Even when Bayesian methods are computationally intractable, they can provide a standard of optimal decision making against which other methods can be measured

Bayes' Theorem: Basics

- Total probability Theorem:

- Bayes' Theorem:

$$p(B) = \sum_i p(B|A_i)p(A_i)$$

$$p(H|X) = \frac{p(X|H)P(H)}{p(X)} \propto p(X|H)P(H)$$

$p(H|X)$: posteriori probability (What we should choose)
 $p(X|H)$: likelihood (What we just see)
 $P(H)$: prior probability (What we knew previously)

Handwritten notes:
 - $p(H|X)$: "Au neg. pos."
 - $p(X|H)$: "the Feature Space window"
 - $P(H)$: "the Feature Space window"

Diagram:
 A grid with columns x_1, x_2, x_3, x_4 and rows y . The grid is labeled $P(X|H)$. A red shaded row is labeled $P(H)$. Below the grid is a blue bar labeled "new data" with a question mark. To the right, $X = \{x_1, x_2, \dots\}$ is shown.

□ **X**: a data sample ("evidence")

□ **H**: X belongs to class C

Prediction can be done based on Bayes' Theorem:

Classification is to derive the maximum posteriori

Naïve Bayes Classifier: Making a Naïve Assumption

- Practical difficulty of Naïve Bayes inference: It requires initial knowledge of many probabilities, which may not be available or involving significant computational cost
- A Naïve Special Case
 - Make an additional assumption to simplify the model, but achieve comparable performance.

Train result $P(x_1, x_2, x_3, x_4 | H)$

$$P(x_1, x_2, x_3, x_4 | H)$$

↓ independent

$$P(x_1 | H) \cdot P(x_2 | H) \cdot \dots$$

$$P(x_4 | H)$$

attributes are conditionally independent
(i.e., no dependence relation between attributes)

$$p(X|C_i) = \prod_k p(x_k|C_i) = p(x_1|C_i) \cdot p(x_2|C_i) \cdot \dots \cdot p(x_n|C_i)$$

$P(\text{Neg.}) > P(\text{Pos.}) \leftarrow$ *Naïve Bayes'* $\leftarrow$

- Only need to count the class distribution w.r.t. features

x_1	x_2	x_3	x_4	y
+	+	-	+	P
-	+	-	-	N
-	+	+	+	P
+	-	+	-	N

new data

$$P(H) = 2/4$$

$$P(x_1 | H) = 1/2$$

$$P(x_2 | H) = 0/2$$

$$P(x_3 | H) = 1/2$$

$$P(x_4 | H) = 0/2$$

$$P(N) = 2/4$$

$$P(x_1 | N) = 1/2$$

$$P(x_2 | N) = 1/2$$

$$P(x_3 | N) = 1/2$$

$$P(x_4 | N) = 2/2$$

Naïve Bayes

① Introduction

- Supervised ML.
- Use for Classification
- Based on Bayes' Theorem

② Foundation

- Bayes' Theorem
- Total probability

$$P(A) = \sum_{i=1}^k P(B_i) P(A|B_i)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- Independence

denoted as $P(A \cap B) = P(A) \cdot P(B)$

$$P(A|B) = P(A)$$

$$P(A_1, A_2, A_3|B) = P(A_1|B) \cdot P(A_2|B) \cdot P(A_3|B)$$

③ Bayes' Theorem

vector of X ; $X = \{x_1, x_2, x_3\}$

$$p(H|X) = \frac{p(X|H)P(H)}{p(X)} \propto p(X|H)P(H)$$

Posteriori Probability Likelihood Prior Probability

! Naïve Bayes' ง่ายกว่า Posteriori Probability ที่จริงที่ถูกต้อง

④ Bayes' → Naïve Bayes'

- Naïve (france, adj) : ไร้เดียงสา, ไร้เดียงสา ...
- Naïve Assumption

To simplify a model, but achieve comparable performance
attributes (X) are conditionally independence

$$p(H|X) = \frac{p(X|H)P(H)}{p(X)} \propto p(X_1|H) \cdot p(X_2|H) \cdot p(X_3|H) \cdot p(X_4|H) \cdot \dots \cdot p(X_k|H) \cdot P(H)$$

$$P(X_1|H) \cdot P(X_2|H) \cdot P(X_3|H) \cdot P(X_4|H) \cdot \dots \cdot P(X_k|H) \cdot P(H)$$

	X	X ₁	X ₂	X ₃	Y
	Animals	Size of Animal	Body Color	Can we Pet them	
0	Dog	Medium	Black	Yes	
1	Dog	Big	White	No	$P(H^-)$
2	Rat	Small	White	Yes	
3	Cow	Big	White	Yes	$P(H^+)$
4	Cow	Small	Brown	No	
5	Cow	Big	Black	Yes	
6	Rat	Big	Brown	No	
7	Dog	Small	Brown	Yes	
8	Dog	Medium	Brown	Yes	
9	Cow	Medium	White	No	
10	Dog	Small	Black	Yes	
11	Rat	Medium	Black	No	
12	Rat	Small	Brown	No	
13	Cow	Big	White	Yes	

$P(X)$

$P(X_3)$

Train data

	X_1	X_2	X_3	Y
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	Animals	Size of Animal	Body Color	Can we Pet them
0	Dog	Medium	Black	Yes H^+
1	Dog	Big	White	No H^-
2	Rat	Small	White	Yes
3	Cow	Big	White	Yes
4	Cow	Small	Brown	No
5	Cow	Big	Black	Yes
6	Rat	Big	Brown	No
7	Dog	Small	Brown	Yes
8	Dog	Medium	Brown	Yes
9	Cow	Medium	White	No
10	Dog	Small	Black	Yes
11	Rat	Medium	Black	No
12	Rat	Small	Brown	No
13	Cow	Big	White	Yes

Test data

Dog	Medium	Black	?
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Test data

Dog	Medium	Black	?
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$$p(H|X) = \frac{p(X|H)P(H)}{p(X)} \propto p(X|H)P(H)$$

① กรณี $P(H^-)$: โจทย์ให้ class เป็น No
Solⁿ

$$P(H^- | X_1, X_2, X_3) = P(X_1 | H^-) \cdot P(X_2 | H^-) \cdot P(X_3 | H^-) \cdot P(H^-)$$

$$= \frac{1}{6} \cdot \frac{2}{6} \cdot \frac{1}{6} \cdot \frac{6}{14}$$

$$= 0.0040$$

② กรณี $P(H^+)$: โจทย์ให้ class เป็น Yes
Solⁿ

$$P(H^+ | X_1, X_2, X_3) = P(X_1 | H^+) \cdot P(X_2 | H^+) \cdot P(X_3 | H^+) \cdot P(H^+)$$

$$= \frac{4}{8} \cdot \frac{2}{8} \cdot \frac{3}{8} \cdot \frac{8}{14}$$

$$= 0.0268$$

$$\because \begin{cases} P(H^- | X) = 0.0040 \\ P(H^+ | X) = 0.0268 \end{cases} \quad \left\{ \begin{array}{l} P(H^+ | X) > P(H^- | X) \end{array} \right.$$

Dog	Medium	Black	Yes
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$n = 6$

	Animals	Size of Animal	Body Color	Can we Pet them
0	Dog	Medium	Black	Yes
1	Dog	Big	White	No
2	Rat	Small	White	Yes
3	Cow	Big	White	Yes
4	Cow	Small	Brown	No
5	Cow	Big	Black	Yes
6	Rat	Big	Brown	No
7	Dog	Small	Brown	Yes
8	Dog	Medium	Brown	Yes
9	Cow	Medium	White	No
10	Dog	Small	Black	Yes
11	Rat	Medium	Black	No
12	Rat	Small	Brown	No
13	Cow	Big	White	Yes

$n = 14$

$$P(H^-) = \frac{6}{14}$$

$n = 8$

	Animals	Size of Animal	Body Color	Can we Pet them
0	Dog	Medium	Black	Yes
1	Dog	Big	White	No
2	Rat	Small	White	Yes
3	Cow	Big	White	Yes
4	Cow	Small	Brown	No
5	Cow	Big	Black	Yes
6	Rat	Big	Brown	No
7	Dog	Small	Brown	Yes
8	Dog	Medium	Brown	Yes
9	Cow	Medium	White	No
10	Dog	Small	Black	Yes
11	Rat	Medium	Black	No
12	Rat	Small	Brown	No
13	Cow	Big	White	Yes

$n = 14$

$$P(H^+) = \frac{8}{14}$$

⑤ Pros & Cons

✓ Pros

- ใช้ง่ายมาก คำนวณการง่าย คำนวณง่าย
- ไม่หนักกับหลายๆ case

✗ Cons

- Assumption ที่ใช้ไม่ได้เสมอไป
- Naïve Assumption

To simplify a model, but achieve comparable performance
attributes (X) are conditionally independence

- Zero probability
ถ้า model พบข้อมูลที่ไม่เคยพบมาก่อน เช่น

Giraffe	Medium	Black	?
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กรณี $P(H^-)$: ข้อมูลมี class เป็น No
Solⁿ

$$\begin{aligned} P(H^- | x_1, x_2, x_3) &= P(x_1 | H^-) \cdot \frac{2}{6} \cdot \frac{1}{6} \cdot \frac{6}{14} \\ &= 0 \cdot \frac{2}{6} \cdot \frac{1}{6} \cdot \frac{6}{14} \\ &= 0 \end{aligned}$$

ทำนองเดียวกันกับ กรณี $P(H^-)$: ข้อมูลมี class เป็น Yes

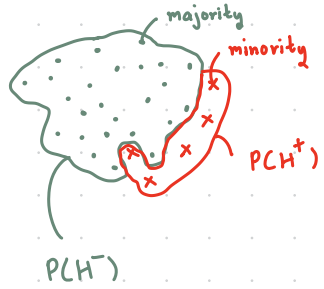
$$\begin{aligned} P(H^- | X) &= 0 \\ P(H^+ | X) &= 0 \end{aligned}$$

ถ้า Laplacian Correction
- +1 ทุก feature case

$$\begin{aligned} P(H^- | x_1, x_2, x_3) &= P(x_1 | H^-)^{+1} \cdot \frac{2+1}{6+3} \cdot \frac{1+1}{6+3} \cdot \frac{6}{14} \\ &= \frac{1}{10} \cdot \frac{3}{9} \cdot \frac{2}{9} \cdot \frac{6}{14} \quad \text{เพราะใน } x_1 \text{ มี Dog Rat Cow Giraffe} \\ &= 0.0032 \end{aligned}$$

$$\begin{aligned} P(H^+ | x_1, x_2, x_3) &= P(x_2 | H^+)^{+1} \cdot \frac{2+1}{8+3} \cdot \frac{3+1}{8+3} \cdot \frac{8}{14} \\ &= \frac{1}{12} \cdot \frac{3}{11} \cdot \frac{4}{11} \cdot \frac{8}{14} \\ &= 0.0047 \end{aligned}$$

⑥ Imbalanced Data



$\therefore$ minimum prior ; $\in$ Constraint

$$\begin{aligned} P(H^- | X) &= 0.0032 \\ P(H^+ | X) &= 0.0047 \end{aligned} \quad \left\{ \begin{array}{l} P(H^+ | X) > P(H^- | X) \end{array} \right.$$

Giraffe

Medium

Black

Yes

Naïve Bayes Classifier: An Example

□ $P(C_i)$: $P(\text{buys_computer} = \text{"yes"}) = 9/14 = 0.643$

$P(\text{buys_computer} = \text{"no"}) = 5/14 = 0.357$

□ Compute $P(X|C_i)$ for each class

$P(\text{age} = \text{"<=30"} | \text{buys_computer} = \text{"yes"}) = 2/9 = 0.222$

$P(\text{age} = \text{"<= 30"} | \text{buys_computer} = \text{"no"}) = 3/5 = 0.6$

$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"yes"}) = 4/9 = 0.444$

$P(\text{income} = \text{"medium"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$

$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$

$P(\text{student} = \text{"yes"} | \text{buys_computer} = \text{"no"}) = 1/5 = 0.2$

$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"yes"}) = 6/9 = 0.667$

$P(\text{credit_rating} = \text{"fair"} | \text{buys_computer} = \text{"no"}) = 2/5 = 0.4$

□ **$X = (\text{age} \leq 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit_rating} = \text{fair})$**

$P(X|C_i)$: $P(X | \text{buys_computer} = \text{"yes"}) = 0.222 \times 0.444 \times 0.667 \times 0.667 = 0.044$

$P(X | \text{buys_computer} = \text{"no"}) = 0.6 \times 0.4 \times 0.2 \times 0.4 = 0.019$

$P(X|C_i) \cdot P(C_i)$: $P(X | \text{buys_computer} = \text{"yes"}) \cdot P(\text{buys_computer} = \text{"yes"}) = 0.028$

$P(X | \text{buys_computer} = \text{"no"}) \cdot P(\text{buys_computer} = \text{"no"}) = 0.007$

Therefore, X belongs to class ("buys_computer = yes")

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Avoiding the Zero-Probability Problem

- Naïve Bayesian prediction requires each conditional probability be **non-zero**

- Otherwise, the predicted probability will be zero

$$p(X|C_i) = \prod_k p(x_k|C_i) = p(x_1|C_i) \cdot p(x_2|C_i) \cdots p(x_n|C_i)$$

- Example. Suppose a dataset with 1000 tuples:

income = low (0), income = medium (990), and income = high (10)

- Use **Laplacian correction** (or Laplacian estimator)

- Adding 1 to each case

ဘာ ၁ ပိုမိုပါသလဲ ၀

$$\text{Prob}(\text{income} = \text{low}) = 1/(1000 + 3)$$

$$\text{Prob}(\text{income} = \text{medium}) = (990 + 1)/(1000 + 3)$$

$$\text{Prob}(\text{income} = \text{high}) = (10 + 1)/(1000 + 3)$$

- The “corrected” probability estimates are close to their “uncorrected” counterparts

Naïve Bayes Classifier: Strength vs. Weakness

- ❑ Strength
 - ❑ Easy to implement
 - ❑ Good results obtained in most of the cases
- ❑ Weakness
 - ❑ Assumption: attributes conditional independence, therefore loss of accuracy
 - ❑ Practically, dependencies exist among variables
 - ❑ E.g., Patients: Profile: age, family history, etc.
Symptoms: fever, cough etc.
Disease: lung cancer, diabetes, etc.
 - ❑ Dependencies among these cannot be modeled by Naïve Bayes Classifier
- ❑ How to deal with these dependencies?
 - ❑ Use Bayesian Belief Networks (to be covered in the next chapter)

